

Parametric curves are functions which define the coordinates of the points in terms of a parameter t

These are represented by two equations

Parametric line:

$$1. \quad qx = (1-t)x(A) + t x(B)$$

$$qy = (1-t)y(A) + t y(B)$$

$$A = (2, 3)$$

$$B = (4, 4)$$

$$qx(t) = (1-t)x(A) + t x(B)$$

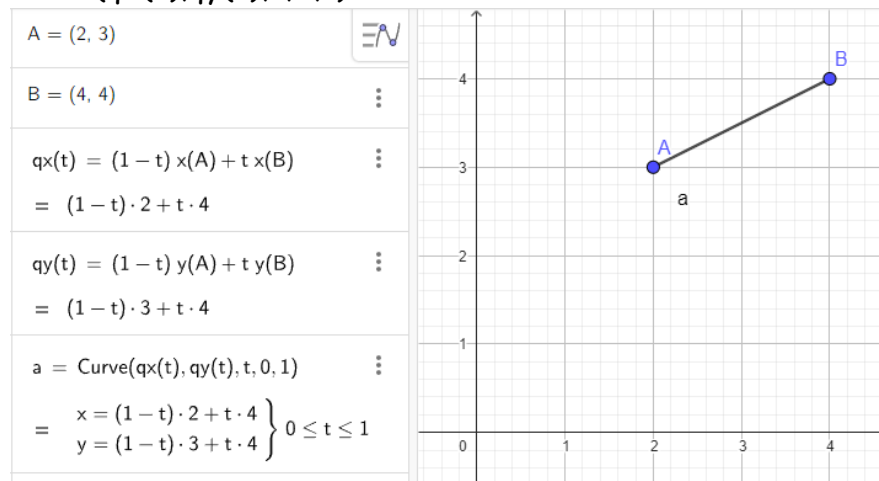
$$= (1-t) \cdot 2 + t \cdot 4$$

$$qy(t) = (1-t)y(A) + t y(B)$$

$$= (1-t) \cdot 3 + t \cdot 4$$

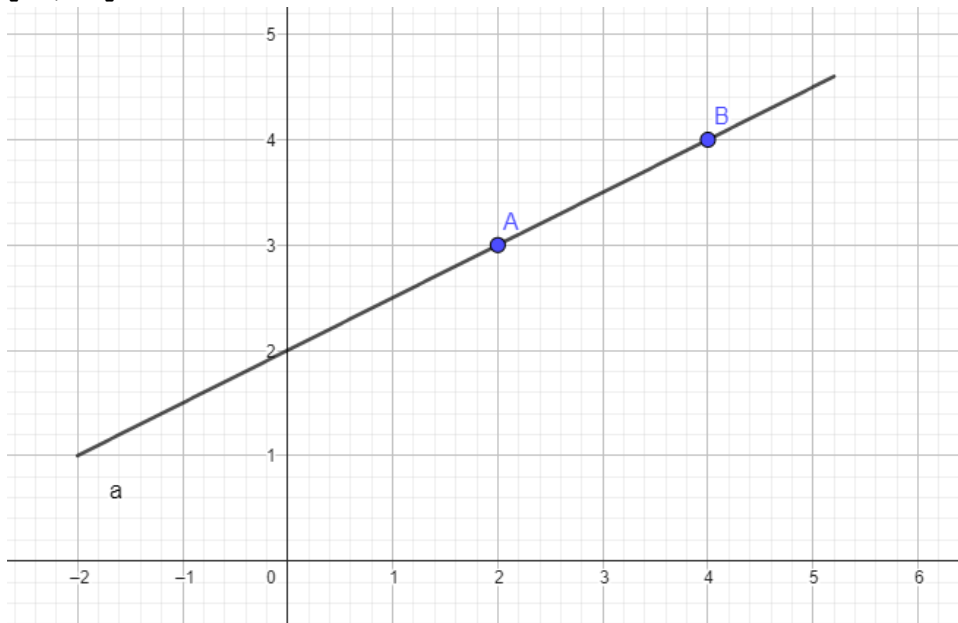
and then we draw the curve using the t parameter between the range of 0 and 1

Curve($qx(t), qy(t), t, 0, 1$)



in this case the starting position is at the A point, we can modify the range to extend the line in each direction

$[-2, 1.6]$:



2. an other way of getting a parametric line is:

taking the vector between B and A

$$\text{vec} = B - A$$

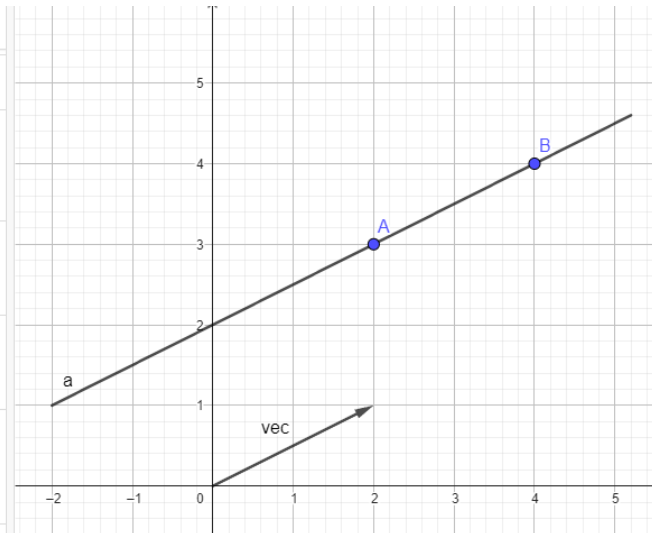
and then creating the functions:

$$nx = x(A) + x(\text{vec}) \cdot t$$

$$ny = y(A) + y(\text{vec}) \cdot t$$

$$\text{Curve}(nx(t), ny(t), t, 0, 1)$$

$A = (2, 3)$	⋮
$B = (4, 4)$	⋮
$\text{vec} = B - A$	⋮
$= \begin{pmatrix} 2 \\ 1 \end{pmatrix}$	⋮
$nx(t) = x(A) + x(\text{vec}) \cdot t$	⋮
$= 2 + 2t$	⋮
$ny(t) = y(A) + y(\text{vec}) \cdot t$	⋮
$= 3 + 1t$	⋮
$a = \text{Curve}(nx(t), ny(t), t, -2, 1.6)$	⋮
$= \left. \begin{array}{l} x = 2 + 2t \\ y = 3 + 1t \end{array} \right\} -2 \leq t \leq 1.6$	



Parametric circle:

radius = 5

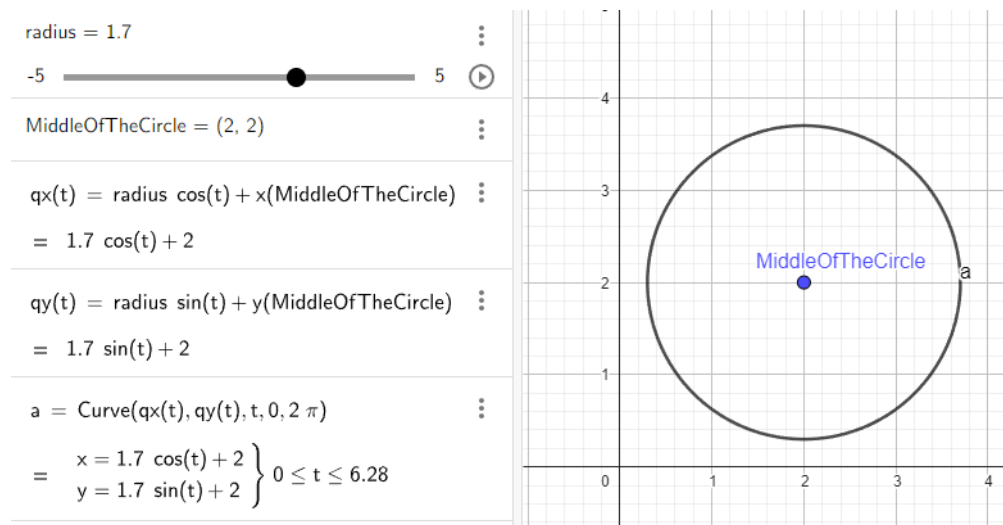
MiddleOfTheCircle = (2,2)

$qx(t) = \text{radius} * \cos(t) + x(\text{MiddleOfTheCircle})$

$qy(t) = \text{radius} * \sin(t) + y(\text{MiddleOfTheCircle})$

a full revolution around the circle is $2*\pi$, so we use that as the range

$\text{Curve}(qx(t), qy(t), t, 0, 2\pi)$



Hermite curve:

$$A = (2,4)$$

$$B = (3,6)$$

$$C = (6,5)$$

$$D = (7,3)$$

we use t_0, t_1, t_2, t_3 parameters, each representing a tangent/slope at a specific point

$$t_0 = -1$$

$$t_1 = 0$$

$$t_2 = 1$$

$$t_3 = 2$$

then we create new points using these tangent values

$$X_0 = (t_0, x(A))$$

$$X_1 = (t_1, x(B))$$

$$X_2 = (t_2, x(C))$$

$$X_3 = (t_3, x(D))$$

Then a function on these new points

$$\text{FitPoly}(X_0, X_1, X_2, X_3)$$

And we do the same with the y coordinates

$$Y_0 = (t_0, y(A))$$

$$Y_1 = (t_1, y(B))$$

$$Y_2 = (t_2, y(C))$$

$$Y_3 = (t_3, y(D))$$

$$\text{FitPoly}(Y_0, Y_1, Y_2, Y_3)$$

We use our t parameters as the range:

Curve($f(t),g(t),t,t0,t3$)

